CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a = 3$ and $b = 4$, the value of $5a^2 - 6b$ is: 1

(a) 20 **(b)** 23 **(c)** 22 **(d)** 21

2. Simplify $5(x + 2) - 3(x - 1)$: 1

(a) $2x + 13$ **(b)** $8x + 13$ **(c)** $2x - 7$ **(d)** $2x + 7$

3. The perimeter of a rectangle with sides $5x + 4$ and $3x - 1$ is: 1

(a) $30x$ **(b)** $8x + 3$ **(c)** $16x + 6$ **(d)** $4x + 10$

4. If $n = 10$, then $\frac{n(n+1)}{2}$ equals: 1

(a) 54 **(b)** 55 **(c)** 57 **(d)** 56

5. Which expression is always even for a whole number n? 1

(a) $n + 2$ **(b)** $2n + 1$ **(c)** $2n + 4$ **(d)** $3n$

6. If $x = 4$ and $y = 5$, the value of $3x - 2y + xy$ is: 1

(a) 21 **(b)** 22 **(c)** 23 **(d)** 24

7. A rule is: output $= 3\times$ input -4. The output for the input 9 is: 1

(a) 24 **(b)** 25 **(c)** 22 **(d)** 23

8. If $x = 6$, the value of $2x^2 + 5$ is: 1

(a) 76 **(b)** 78 **(c)** 79 **(d)** 77

9. **Assertion (A):** 'Five more than three times n' is written as 3n + 5. 1
Reason (R): Three times n is written as 3n.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The value of 2x + 3 when x = 4 is 11. 1
Reason (R): In 2x, the number 2 and the letter x are multiplied.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Simplify: $7(x-2)-3(x-5)$. 2

12. Write the sum of three consecutive multiples of 2, starting with $2n$, and simplify it. 2

13. Find the value of $2x^2-3x+4$ when $x=5$. 2

14. Simplify: $9(x-3)-4(x-2)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangle has length $(5x+4)$ cm and breadth $(3x-1)$ cm. Find its perimeter as an expression and its value for $x=6$. 3

16. Write the sum of 3 consecutive numbers starting at n as an expression in n, and evaluate it for $n=7$. 3

17. Taxi A charges ₹20 plus ₹15 per km and taxi B charges ₹60 plus ₹10 per km. Write both fares for k km. For a journey of 6 km, which taxi is cheaper and by how much? 3

OR

A rectangle has length $(7x+3)$ cm and breadth $(2x-1)$ cm. Find its perimeter as an expression and its value for $x=5$. 3

18. Write the sum of 4 consecutive numbers starting at n as an expression in n, and evaluate it for $n=5$. 3

OR

Taxi A charges ₹30 plus ₹12 per km and taxi B charges ₹90 plus ₹7 per km. Write both fares for k km. For a journey of 15 km, which taxi is cheaper and by how much? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sunita makes a row of triangles with matchsticks. One triangle needs 3 sticks, two triangles joined in a row need 5 sticks, and three triangles need 7 sticks. 5

(i) How many sticks are needed for 5 triangles?

(ii) Write an expression for the sticks for n triangles.

(iii) Find the sticks for 10 triangles.

(iv) She has 41 sticks. How many triangles can she make?

20. In a pattern, figure n has $5n+2$ tiles. Find 5

(i) the tiles in figure 12,

(ii) the total tiles in figures 1 and 2, and

(iii) how many more tiles figure 12 has than figure 11.

OR

Priya has ₹x. Her brother has ₹20 more than twice her money. Write expressions for the brother's money and for their total money, and find both when $x=35$. 5

END OF QUESTION PAPER — Total: 40 Marks